MIL-T-46335A(MU)
27 December 1971
SUPERSEDING
MIL-T-46335 (Ord)
20 April 1962

MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M100, M100C and M100D

1. SCOPE

1.1 This specification covers one basic type of Telescope known as Telescope, Panoramic M100, (see 6.1). The additional designations identify certain application differences, and reticle design.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging of
MIL-I-45607	Inspection Euipment, Supply and Maintenance, of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army, Frankford Arsenal

F7695888	Telescope, Panoramic: M100, M100D
F8270977	Telescope, Panoramic: M100C

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PACKAGING DATA SHEETS

7695888	Packaging of Telescope, Panoramic, M100
8270977	Packaging of Telescope, Panoramic, M100C
8214000	Packaging of Telescope, Panoramic, M100D

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contacting officer.)

3. REQUIREMENTS

3.1 Fabricaiton. - The Telescope shall be manufactured in accordance with drawing applicable for specified model number (see 2.1) and drawings pertaining thereto.

3.2 General specifications. - The contractor shall be responsible for adherence to and compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.2.1 MIL-O-13830.

(a) Cleanliness
(b) Illuminated reticle
(c) Resolution

3.2.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Part identification and marking
(f) Workmanship

3.3 Environmental.

3.3.1 Storage temperature. - The telescope shall be exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit. Upon return to standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) the telescope shall show no indication of physical failure and shall meet the performance requirements of this specification.

3.3.2 Operating temperature. - The telescope shall not exceed the maximum allowable torque values of 3.8 when exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.3.3 Vibratinon. With the telescope mounted in its normal operating position, the telescope shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds.

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Subsequent to vibration the telescope shall show no evidence of loose or damaged parts and shall meet the requirements of 3.4 to 3.10 inclusive.

3.4 Cleanliness.

3.4.1 Dirt. - Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 roils diameter shall be three(3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.4.2 Other defects. - There shall be no evidence of moisture, grease, finger print, condensates, fractures and adhesive separations when viewing through the objective and eye end of the instrument.

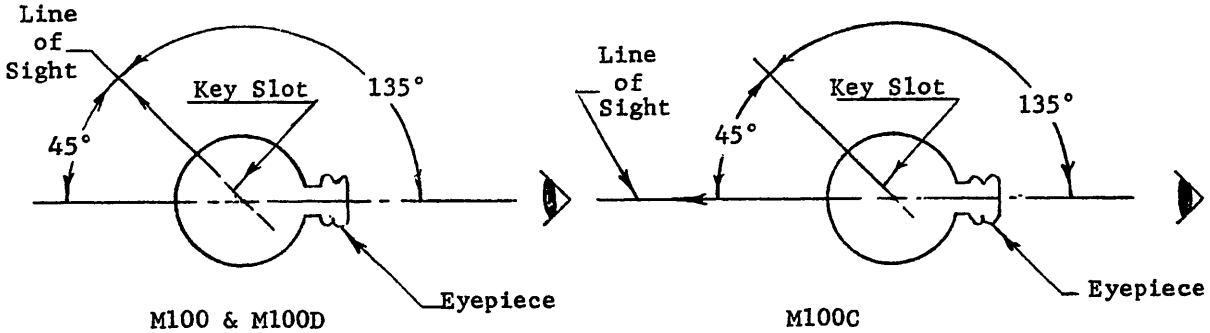
3.5 Sealing and Purging.

3.5.1 Sealing. - The telescope when subjected to an internal pressure of five (5) pounds per square inch gage pressure (psig) of dry nitrogen shall show no evidence of leakage in excess of 0.20 psig over a period of two (2) hours. The dew point of nitrogen shall be at least as low as minus 25 degrees Fahrenheit.

3.5.2 Purging. - The telescope shall be flushed with dry nitrogen supplied at a pressure of 5 ± 1/4 psig until the dew point of the emergent gas is no higher than -25° Fahrenheit. Purging apertures shall be quickly and positively closed to minimize intermingling of external atmosphere upon disconnection of purging equipment.

3.6 Accuracy.

3.6.1 Collimation. - With the azimuth scale indicating zero, the position of the line of sight with reference to the locating key slot shall be within plus or minus 5.0 roils of the position indicated. (See illustratins).



M100 & M100D
Azimuth Scale at Zero

M100C
Azimuth Scale at Zero

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3.6.2 Image Tilt. - With the axis of the telescope vertical, as defined by the mounting surfaces, the image of a vertical target line shall be vertical within one (1) degree of arc throughout two (2) revolutions of the head.

3.6.3 Plumb travel. - With the telescope positioned as in 3.6.2 and the telescope vertical, the line of sight shall follow a plumb line (starting with the line of sight at zero elevation and in coincidence with the plumb line) within 0.5 mil (total spread of 1.0 mil) when depressed a minimum of 300 mils and elevated 300 mils from zero elevation at any azimuth setting through at least two full revolutions of the head.

3.6.4 Parallelism of reticle and image. - With the telescope positioned as in 3.6.2 the vertical reticle line shall be parallel to the image of a vertical target line within 30 minutes of arc at any azimuth setting over two full revolutions of the head.

3.6.5 Level travel. - With the telescope positioned as in 3.6.2 (the line of sight horizontal) the line of sight shall not deviate in elevation from a horizontal target line by more than 5 mils (total excursion of 10 mils) when the head is rotated through at least two full revolutions.

3.6.5.1 Lift. - Vertical displacement of the line of sight due to reversing direction of rotation of the head shall not exceed 1.0 mil.

3.6.6 Azimuth error. - Error at azimuth readings in 800 mil increments throughout two full revolutions of the head in each direction shall not exceed 1.5 mil (including backlash) as read on the coarse and fine scales. The total spread between the greatest positive error and the greatest negative error shall not exceed 2.0 mils The total error including backlash between scale and counter reading shall not exceed 1.0 mil.

3.6.6.1 Backlash. - Backlash in the azimuth mechanism shall not exceed 1.0 mil.

3.6.7 Eyepiece focus. - Eyepiece focus shall be fixed between the limits of minus 0.75 and minus 1.0 diopter.

3.6.8 Resolution. - Resolution on the optical axis shall be 12 seconds of arc or less when using an observation telescope of at least 3 power and at least 0.4 inch entrance pupil. Resolution consists of detectable line structure and proper line count in all four meridians. Total spread in focus shift between meridians shall not exceed 1/4 diopter measured at the instrument eyepiece.

3.6.9 Parallax. - Parallax at the center of the field shall not exceed 0.15 mil when viewing a target at 80 meters ± meter.

3.6.10 Reticle Accuracy. - The angular value between extreme reticle graduations shall be as indicated on the reticle drawing within three (3) percent.

3.7 Counter operation. - The counter shall be resettable to 3200 mils plus or minus 0.25 mil from any reading. The counter shall follow properly when the operating knob is turned at a rate of six revolutions per second for a period of one minute.

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3.8 Operability. - The running torque or force required to actuate the following knobs in each direction at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) shall not exceed the values specified below:

Azimuth knob	10 inch pounds
Counter reset knob (rotation)	5 inch pounds
Counter reset knob (push in to reset)	7.5 pounds
Elevation knob	6 inch pounds

The values obtained when exposed to the extreme temperatures of 3.3.2 shall not exceed the value recorded at the standard ambient temperature by more than 100 percent.

3.9 Illumination. - With the lamps for illumination of the azimuth scale and counter and reticle energized with 3 volts the markings shall be clearly defined when observed in a darkened are a.

3.10 Electrical.

3.10.1 Insulation resistance. - Insulation resistance shall not be less than 10 megohms with a potential of 100 volts direct current applied.

3.10.2 Continuity. - The continuity of the wiring shall be checked for conformance with the applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspeciton. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Excpt as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specificaiton. Definitions-of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.32.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence fo SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-1O5 and the sampling plans specified in Table I herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Sectin 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

Use 100% Inspection

Characteristic		Requirement	Test Procedures
101.	Physical failure, post vibration	3.3.3	4.7.1
102.	Cleanliness	3.4	4.7.2
103.	Sealing	3.5	4.7.3
104.	Collimation	3.6.1	4.7.4.1
105.	Image tilt	3.6.2	4.7.4.2
106.	Plumb travel	3.6.3	4.7.4.3.1
107.	Reticle cant.	3.6.4	4.7.4.3.2
108.	Level travel	3.6.5	4.7.4.3.3
109.	Lift	3.6.5.1	4.7.4.3.4
110.	Azimuth error	3.6.6	4.7.4.3.5

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Characteristic	Requirement	Test Procedures
111. Backlash	3.6.6.1	4.7.4.3.6
112. Eyepiece focus	3.6.7	4.7.4.4
113. Resolution	3.6.8	4.7.4.5
114. Parallax	3.6.9	4.7.4.6
115. Reticle accuracy	3.6.10	4.7.4.7
116. Counter malfunction	3.7	4.7.4.8
117. Control knobs, torque	3.8	4.7.5
118. Reset knob, force	3.8	4.7.5
119. Illumination (Scale and Counter)	3.9	4.7.6
120. Illumination (Reticle)	3.9	4.7.6
121. Insulation Resistance	3.10.1	4.7.7
122. Continuity	3.10.2	4.7.8

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling for environmental testing.

4.4.1 Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STS-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examinations and tests. The requirements of 3.1 and 3.2 shall be accepted on contractor's certification.

TABLE II

Characteristic	Requirement	Test Procedure
301. Temperature (Storage)	3.3*	4.6.1
302. Temperature (Operating)	3.3.	4.6.2
303. Torque (Operating Temperature)	3.3.2	4.7.5
304. Reset Counter Force (Operating Temperature)	3.3.2	4.7.5

4.4.2 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607. I

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate accept-ability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and 4.7.

NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926
2. Continuity checker	Standard shop type capable of providing readings accurately within 2% of specified tolerance
3. Insulation resistance checker	Standard shop type conforming to the accuracies outlined in 4.5.2.2
4. Power supply	0 to 5 vdc output within 10% accuracy
5. Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Storage temperature. - The temperature cycling for the physical endurance test of the telescope at minus 80 degrees and plus 160 degrees

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Fahrenheit shall be conducted in accordance with Procedure II of Specification MIL-F-13926. Upon return to standard ambient temperature, the telescope shall be subjected to a visual and tactile examination and shall show no evidence of electrical, mechanical or optical failure. The telescope shall then be sujected to the tests specified herein, and shall conform to the requirements specified. The storage temperature test shall be performed on the production model only.

4.6.1.2 Operating temperature. - The temperature cycling for the operability test of the telescope at the specified temperature extremes shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond minus 40 degrees Fahrenheit or raised beyond plus 150 degrees Fahenheit. When the telescope is thermally stabilized at minus 40 degrees and plus 150 degrees Fahrenheit, respectively, the indicated control knobs shall be subjected to the tests specified in 4.7.5. The control knobs shall conform to the operability requirements for each temperature extreme- The torque and force values obtained at each temperature extreme shall be recorded for comparison with those values obtained at standard ambient temperature.

4.7 Performance Testing.

4.7.1 Vibratin. - The telescope shall be vibrated (on a device conforming to the requirements of drawing F7560085) in accordance with Vibration Procedure V of Specification MIL-F-13926, except that the duration shall be five minutes plus or minus fifteen seconds. Upon completion of the vibration operation the telescope shall be subjected to a visual and tactile examination and shall show no evidence of electrical, mechanical or optical failure. The telescope shall then be subjected to the tests specified in 4.7.2 through 4.7.8 inclusive, and shall conform to the requirements specified.

4.7.2 Cleanliness. - The optical surfaces of the completed telescope shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet the requiements specified. Dirt particles shall not exceed the limit of 3.4.

4.7.3 Sealing. - This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy depicted on Drawing F8565556. When using pressure test fixture, the fixture shall be set-up as outlined in set-up instructions shown on Drawing F8565556, Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the panoramic telescope, adjust testing pressure to the pressure specified in 3.5.1 and maintain the pressure for time specified in 3.5.1. Pressure gage shall remain within 0.2 to assure compliance with 3.5.1. At the completion of this test, connect a dew point tester to the exit port and purge the optical assembly until -25°F is recorded and seal the instrument.

4.7.4 Accuracy. - All accuracy tests of the panoramic telescope shall be conducted with a calibrated target positioned at the specified minimum parallax distance, except as otherwise specified.

4.7.4.1 Collimation. - This test shall be conducted with the telescope positioned on Test Fixture 7695248 as specified on Drawing F7695248. Position

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the telescope reticle coincident with the collimator reticle image. The azimuth scale shall be within the specified tolerance of its zero position.

4.7.4.2 Image tilt. - This test of the telescope shall be conducted in accordance with the set-up and operating instructions on Drawing D7695253 using Test Fixture 769523. The mounting surfaces of the panoramic telescope shall be used in establishing its vertical position. The image of the plumb line formed by the optical system of the panoramic telescope shall be vertical within the tolerance specified as measured on the calibrated reticle of the auxiliary telescope.

4.7.4.3 Accuracy. - The following tests of the telescope shall be conducted in accordance with the set-up and operating instructions on Drawing F7695120 using Test Fixture 7695120.

4.7.4.3.1 Plumb travel. - Establish the telescope line of sight through the intersection of the reticle lines horizontal and coincident to the plumb line target image. Elevate and depress the line of sight 300 mils from the horizontal by means of the elevation knob. Any deviation of the line of sight from the plumb line image during each excursion-from the horizontal shall not exceed the tolerance specified. This test shall be performed at eight azimuth settings equally spaced through a minimum of two revolutions of the telescope head (12,800 mils).

4.7.4.3.2 Parallelism of reticle and image. - Position one extremity of a 70 mils section of the telescope vertical reticle line coincident to a plumb line targe image. The opposite extremity of the 70 mils section of the reticle line shall be coincident to the plumb line image within the tolerance specified, as measured on the target. This test shall be performed at eight azimuth settings equally spaced through a minimum of two revolutions of the telescope head.

4.7.4.3.3 level travel. - Establish the telescope line of sight through the intersection of the reticle lines horizontal and coincident to the horizontal target line image. Rotate the telescope line of sight through a minimum of two revolutions of the head assembly (12,800 mils by means of the azimuth knob. During this excursion any vertical deviation of the line of sight from the horizontal target line shall not exceed the tolerance specified, as measured on the target. Level travel shall be tested at a minimum of eight azimuth settings equally spaced through a minimum of two revolutions of the telescope head.

4.7.4.3.4 Lift (deflection). - This test will be conducted concurrent to the level travel test. During level travel, the direction of head rotation shall be reversed. Any vertical displacement of the line of sight with respect to the horizontal target line resulting from this reversing of travel shall not exceed the tolerance specified, as measured on the target. This test shall be performed at a minimum of eight azimuth settings equally-spaced through a minimum of two revolutions of the telescope head.

4.7.4.2.5 Azimuth error. - Position the telescope vertical reticle line coincident to the collimator plumb line image with backlash removed for direction of travel. Turn the collimator clockwise through 800 mils then

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with the azimuth knob, rotate the telescope head clockwise and obtain coincidence of the boresight point (intersection of the reticle lines) and the plumb lin image withour over-tRecord azimuth reading of scales. The azimuth scales shall read 800 mils plus or minus 1.5 mils. Rotate the azimuth knob one full revolution clockwise, then with backlash included, rotate the knob counterclockwise and obtain coincidence of the boresight point and plumb line image without overtravel. The second recorded reading shall be 800 mils plus or minus 1.5 roils on the azimuth scale. Any error between the mil value on the scale and counter for the first or second recorded reading shall not exceed 1.0 mil. Rotate the testing device to 1600 mils and test the telescope for azimuth accuracy by application of the above procedure. The azimuth accuracy of the telescope shall be tested in 800. mil increments throughout two revolutions of the head clockwise then counterclockwise. The total spread between the greatest positive error and the greatest negative error obtained in the remainder from any two recorded readings throughout two revolutions of the head clockwise or two revolutions of the head counterclockwise shall not exceed 2.0 mils.

4.7.4.3.6 Backlash. - The backlash test in the azimuth mechanism shall be conducted concurrent with the test specified in 4.8.7.3.5. Backlash shall not exceed the tolerance specified.

4.7.4.4 Eyepiece focus. - This test shall be performed with the aid of a calibrated dioptometer having a magnification of at least three power. Adjust the dioptometer for infinity focus for the observer’s eye with the diopter measuring scale set to zero. The telescope reticle at the center of the field shall be in best focus within the limits specified as measured on the diopter scale of the dioptometer.

4.7.4.5 Resolution. - This test shall be conducted in accordance with Specification MIL-O-13830. The resolving power of the telescope shall meet the requirements specified.

4.7.4.6 Parallax. - Parallax in the center of the field (geometric center of the reticle not exceed the tolerance specified while viewing targets positioned the specified distances from the telescope. Optically, parallax in the telescope is any apparent movement of the reticle in relation to distance targets seen within the field of view caused by lateral movement of the observer's eye. For purpose of this test, the thickness of the telescope reticle line may be considered as equal to 0.16 mil.

4.7.4.7 Reticle accuracy. - The test for reticle accuracy shall be conducted in accordance with Specification MIL-O-13830. The angular subtense of the reticle graduations shall not exceed the tolerance specified as measured on the target.

4.7.4.8 Counter operation. - At a minimum of four azimuth scale settings, equally spaced throughout 6400 mils, depress and rotate the counter reset knob and se 3200 mils on the counter. Upon release of the reset knob the counter shall remain at 3200 roils within the tolerance specified. Rotate the azimuth knob in a clockwise and counterclockwise direction for one minute at the rate of 6 revolutions per second. The excursion in each direction shall cover at

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least 3200 mils as indicated on the counter. During the excursions the counter shall correctly follow the azimuth scale.

4.7.5 Operability. - The force and running torques required to operate the specified control knobs shall be measured by use of standard measuring equipment that is capable of measuring the applied force and torques. Force and torque values obtained at minus 40 degrees and plus 150 degrees Fahrenheit shall be recorded for comparison with the values obtained at standard ambient temperature. The maximum force and running torque required to operate the specified control knobs shall not exceed the value specified for the temperature indicated.

4.7.6 Illumination. - This test shall be performed in a darkened area. When the lamps are energized with the indicated voltage the scale and counter numerals shall be clearly defined. The reticle pattern shall be clearly defined when visually inspected against the dark background. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall be cause for rejection.

4.7.7 Insulation resistance. - This test shall be conducted in accordance with Method 302 of Standard MIL-STD-202 using test condition "A". Insulation resistance measurements shall be made between insulated points and ground. Insulation resistance shall not be less than that specified in 3.10.1.

4.7.8 Continuity. - The electrical circuits shall be tested for continuity by use of standard measuring equipment and shall conform to the wiring diagram.

4.8 Packaging, packing and marking. - Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - The telescope shall be packaged, packed and marked in accordance with applicable packaging data sheet (see 2.1). The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Panoramic Telescope: M100 and M100D mounts in Telescope Mounts of the M99 Series for laying the full tracked, 105mm, M52 and 8 inch, M55 Self Propelled Howitzers in azimuth for indirect fire. The Panoramic Telescope: M100C is intended to lay the 105MM armament of the Self Propelled Howitzer, LTV-A6 (Navy) for indirect fire.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 2.1).
(e) Provisions for first article testing.

Custodian:
Army - MU

Preparing activity:
Army - MU
Project No. 1240-A338

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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